Haohong Zhang.

Bioinformatics / Foundation models / AI for biology

Ph.D. student · Huazhong University of Science and Technology · Wuhan, China

haohongzh@gmail.com · ludenszhang.github.io · GitHub · Scholar · ORCID 0000-0001-6267-4244

I develop foundation models and interpretable machine learning methods for microbiome research, connecting microbial communities, genome representations and multi-omics data with biological questions.

01 Education

2023 - Present **Ph.D. in Bioinformatics** · Direct-track

Huazhong University of Science and Technology · Advisor: Prof. Kang Ning

2019 - 2023 **Bachelor of Science**

Huazhong University of Science and Technology · Dengfeng Program, Class of 1901

02 Technical skills

Machine learning. Python, PyTorch, PyTorch Lightning, Hugging Face Transformers and Datasets; masked language modeling, transfer learning and autoencoders.

Biological data. Biopython, FASTA processing, genome dataset preparation, microbiome abundance profiles and discrete sequence representations.

Analysis & tooling. NumPy, pandas, SciPy, scikit-learn, scikit-bio, plotnine; PCA/PCoA, command-line tools, Python packaging and TensorBoard.

03 Research software

MGM2 Repository

A multimodal foundation model for microbial community states and responses, released in four pretrained sizes.

MGM Repository

Large-scale pretraining for transferable microbiome analysis across diverse contexts.

MicroVQVAE Repository

A genome foundation model that learns discrete, context-aware genome tokens from ordered prokaryotic protein sequences.

EXPERT-lightning Repository

Context-aware microbial source tracking with transfer learning and biome ontology.

DeepMicroCancer Repository

Microbiome-based models for the diagnosis of a broad spectrum of cancer types.

ASD-cancer Repository

Interpretable survival subtyping from microbial profiles and host gene expression.

04 Selected peer-reviewed publications

MGM as a Large-Scale Pretrained Foundation Model for Microbiome Analyses in Diverse Contexts

H Zhang, Y Zhang, Z Kang, J Xiong, R Yang, K Ning · Advanced Science 13(24), e13333 (2026). doi:10.1002/advs.202513333

Large-scale pretraining / Microbiome / Transfer learning

Foundation Models for Microbiome Research: From Sequence Semantics to Community Dynamics and Multimodal World Models

H Zhang, Z Kang, K Ning · Advanced Genetics 7(3), e70046 (2026). doi:10.1002/ggn2.70046

Foundation models / Microbiome / Multimodal world models

mHolmes improves cross anatomical cadaveric microbiome forecasting for postmortem interval estimation

J Han, Y Zhang, **H Zhang**, K Zhou, X Chen, K Ning · Nature Communications (2026). doi:10.1038/s41467-026-77510-3

Microbiome forecasting / Postmortem interval / Transfer learning

Deep learning enabled integration of tumor microenvironment microbial profiles and host gene expressions for interpretable survival subtyping in diverse types of cancers

H Zhang, X Xiong, M Cheng, L Ji, K Ning · mSystems 9(12), e01395-24 (2024). doi:10.1128/msystems.01395-24

Multi-omics integration / Cancer survival / Interpretable deep learning

Tracing human life trajectory using gut microbial communities by context-aware deep learning

H Zhang, H Chong, Q Yu, Y Zha, M Cheng, K Ning · Briefings in Bioinformatics 24(1), bbac629 (2023). doi:10.1093/bib/bbac629

Life trajectories / Gut microbiome / Context-aware learning

05 Preprints

MGM2 as a Multimodal Foundation Model for Microbial Community States and Responses

H Zhang, Y Zhang, Y Qi, T Liu, L Zhao, R Yang, K Ning · bioRxiv (2026). doi:10.64898/2026.07.20.739063

Foundation models / Multimodal learning / Community responses

MetaClaw: an auditable AI agent for end-to-end, multi-directional metagenomic and multi-omics analysis

H Zhang, Z Li, P N P Lagniton, Z Wang, L Zhao, W Li, P Duan, X Jiang, K Ning · bioRxiv (2026). doi:10.64898/2026.07.21.739769

AI agents / Metagenomics / Reproducible workflows

06 Conference presentation

A discrete token language of prokaryotic genomes learned by MicroVQVAE

Haohong Zhang, Kang Ning · Poster, Cold Spring Harbor Asia: AI and Biology.
April 20-24, 2026 · Suzhou, China.

Conference program · Poster 41

07 Honors

Outstanding Graduate, 2023 · Huazhong University of Science and Technology